

Business Case: Aerofit - Descriptive Statistics & Probability

Mindset:

1. Evaluation will be kept lenient, so make sure you attempt this case study.
2. It is understandable that you might struggle with getting started on this. Just brainstorm, discuss with peers, or get help from TAs.
3. There is no right or wrong answer. We have to get used to dealing with uncertainty in business. This is exactly the skill we want to develop.

About Aerofit

Aerofit is a leading brand in the field of fitness equipment. Aerofit provides a product range including machines such as treadmills, exercise bikes, gym equipment, and fitness accessories to cater to the needs of all categories of people.

Business Problem

The market research team at AeroFit wants to identify the characteristics of the target audience for each type of treadmill offered by the company, to provide a better recommendation of the treadmills to the new customers. The team decides to investigate whether there are differences across the product with respect to customer characteristics.

1. Perform descriptive analytics **to create a customer profile** for each AeroFit treadmill product by developing appropriate tables and charts.
2. For each AeroFit treadmill product, construct **two-way contingency tables** and compute all **conditional and marginal probabilities** along with their insights/impact on the business.

Dataset

The company collected the data on individuals who purchased a treadmill from the AeroFit stores during the prior three months. The dataset has the following features:

Dataset link: **Aerofit_treadmill.csv**

(https://d2beiqkhq929f0.cloudfront.net/public_assets/assets/000/001/125/original/aerofit_treadr1639992749)

Product Purchased:	KP281, KP481, or KP781
Age:	In years
Gender:	Male/Female
Education:	In years
MaritalStatus:	Single or partnered
Usage:	The average number of times the customer plans to use the treadmill each week.
Income:	Annual income (in \$)
Fitness:	Self-rated fitness on a 1-to-5 scale, where 1 is the poor shape and 5 is the excellent
Miles:	The average number of miles the customer expects to walk/run each week

Product Portfolio:

- The KP281 is an entry-level treadmill that sells for \$1,500.
- The KP481 is for mid-level runners that sell for \$1,750.

- The KP781 treadmill is having advanced features that sell for \$2,500.

What good looks like?

1. Import the dataset and do usual data analysis steps like checking the structure & characteristics of the dataset
2. Detect Outliers (using boxplot, “describe” method by checking the difference between mean and median)
3. Check if features like marital status, age have any effect on the product purchased (using countplot, histplots, boxplots etc)
4. Representing the marginal probability like - what percent of customers have purchased KP281, KP481, or KP781 in a table (*can use `pandas.crosstab` here*)
5. Check correlation among different factors using heat maps or pair plots.
6. With all the above steps you can answer questions like: What is the probability of a male customer buying a KP781 treadmill?
7. **Customer Profiling** - Categorization of users.
8. **Probability**- marginal, conditional probability.
9. Some recommendations and actionable insights, based on the inferences.

Later on, we will see more ways to do “customer segmentation”, but this case study in itself is relevant in some real-world scenarios.

Evaluation Criteria

1. Defining Problem Statement and Analysing basic metrics **(10 Points)**
 1. Observations on shape of data, data types of all the attributes, conversion of categorical attributes to 'category' (If required), statistical summary
2. Non-Graphical Analysis: Value counts and unique attributes **(10 Points)**
3. Visual Analysis - Univariate & Bivariate **(30 Points)**
 1. For continuous variable(s): Distplot, countplot, histogram for univariate analysis (10 Points)
 2. For categorical variable(s): Boxplot (10 Points)
 3. For correlation: Heatmaps, Pairplots(10 Points)
4. Missing Value & Outlier Detection **(10 Points)**
5. Business Insights based on Non-Graphical and Visual Analysis **(10 Points)**
 1. Comments on the range of attributes
 2. Comments on the distribution of the variables and relationship between them
 3. Comments for each univariate and bivariate plot
6. Recommendations **(10 Points)** - Actionable items for business. No technical jargon. No complications. Simple action items that everyone can understand

Submission Process:

- Type your insights and recommendations in the text editor.
- Convert your jupyter notebook into PDF (Save as PDF using Chrome browser’s Print command), upload it in your Google Drive (set the permission to allow public access), and paste that link in the text editor.
- Optionally, you may add images/graphs in the text editor by taking screenshots or saving matplotlib graphs using `plt.savefig(...)`.
- After submitting, you will not be allowed to edit your submission.

#LINK TO THIS COLAB: <https://colab.research.google.com/drive/1PLgWJ-g23dm6r-j4-ixRGLEF3BzqkABP?usp=sharing>

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
url="https://d2beiqrkhq929f0.cloudfront.net/public_assets/assets/000/001/125/original/aerofit_treadmill.csv?163999"
```

```
df=pd.read_csv(url)
df.head() # show the first n rows of the dataset
```

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
0	KP281	18	Male	14	Single	3	4	29562	112
1	KP281	19	Male	15	Single	2	3	31836	75
2	KP281	19	Female	14	Partnered	4	3	30699	66
3	KP281	19	Male	12	Single	3	3	32973	85
4	KP281	20	Male	13	Partnered	4	2	35247	47

```
df.tail()
```

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
175	KP781	40	Male	21	Single	6	5	83416	200
176	KP781	42	Male	18	Single	5	4	89641	200
177	KP781	45	Male	16	Single	5	5	90886	160
178	KP781	47	Male	18	Partnered	4	5	104581	120
179	KP781	48	Male	18	Partnered	4	5	95508	180

```
df.shape # size od dataset
```

```
(180, 9)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 180 entries, 0 to 179
Data columns (total 9 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Product         180 non-null   object
1   Age             180 non-null   int64
2   Gender          180 non-null   object
3   Education       180 non-null   int64
4   MaritalStatus   180 non-null   object
5   Usage          180 non-null   int64
6   Fitness         180 non-null   int64
7   Income          180 non-null   int64
8   Miles          180 non-null   int64
dtypes: int64(6), object(3)
memory usage: 12.8+ KB
```

```
df.describe()
```

	Age	Education	Usage	Fitness	Income	Miles
count	180.000000	180.000000	180.000000	180.000000	180.000000	180.000000
mean	28.788889	15.572222	3.455556	3.311111	53719.577778	103.194444
std	6.943498	1.617055	1.084797	0.958869	16506.684226	51.863605
min	18.000000	12.000000	2.000000	1.000000	29562.000000	21.000000
25%	24.000000	14.000000	3.000000	3.000000	44058.750000	66.000000
50%	26.000000	16.000000	3.000000	3.000000	50596.500000	94.000000
75%	33.000000	16.000000	4.000000	4.000000	58668.000000	114.750000
max	50.000000	21.000000	7.000000	5.000000	104581.000000	360.000000

```
df.isnull().sum()
```

	0
Product	0
Age	0
Gender	0
Education	0
MaritalStatus	0
Usage	0
Fitness	0
Income	0
Miles	0

dtype: int64

df["Product"].value_counts()

What is the best selling product? -> KP281

	count
Product	
KP281	80
KP481	60
KP781	40

dtype: int64

Are more males or females buying treadmills? Males
df["Gender"].value_counts()

	count
Gender	
Male	104
Female	76

dtype: int64

who buys more single, married people
df["MaritalStatus"].value_counts()

	count
MaritalStatus	
Partnered	107
Single	73

dtype: int64

Univariate Analysis

Age

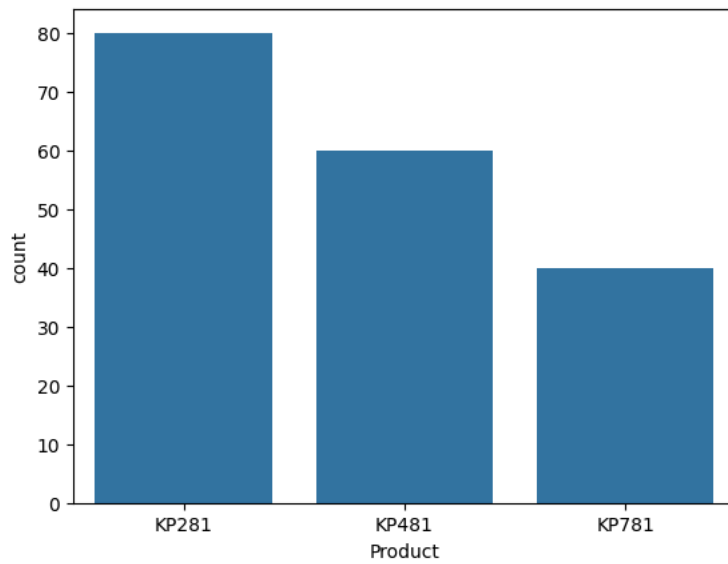
"""

- What is the average age of people?
- Which age group buys most treadmills?
- What is the min and max age?

"""

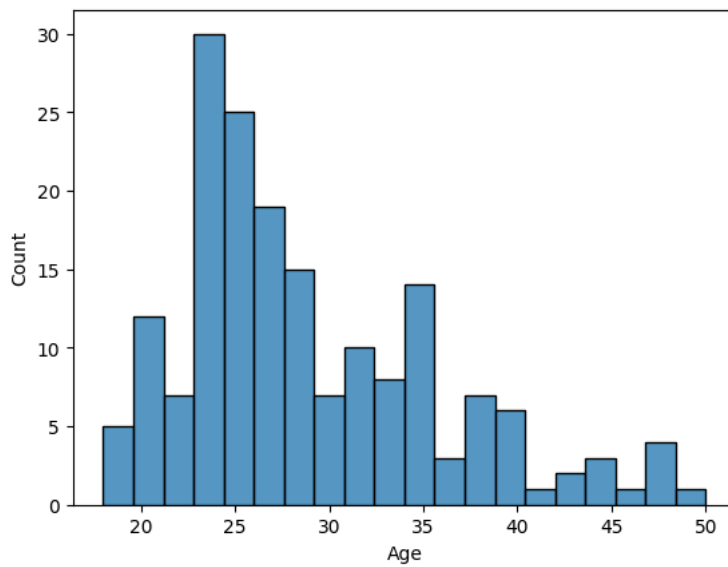
sns.countplot(x=column,data=df)
sns.countplot(x="Product",data=df)
plt.show()

KP281 is the most popular it might be an entry level product



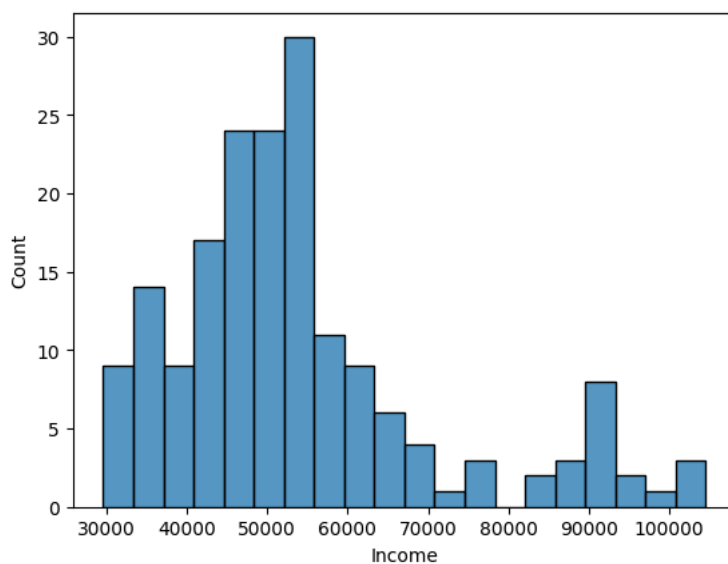
```
sns.histplot(df["Age"],bins=20)  
plt.show()
```

Most buyers are from the age group 25 – 30 years



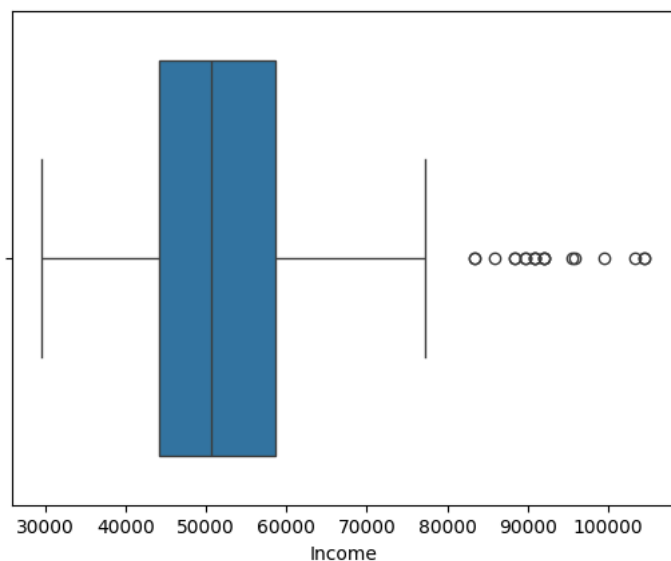
income distribution

```
sns.histplot(df["Income"], bins=20)  
plt.show()
```



```
# outlier detection

sns.boxplot(x=df["Income"])
plt.show()
```



```
print("Mean Income", df["Income"].mean())
print("Median Income",df["Income"].median())
```

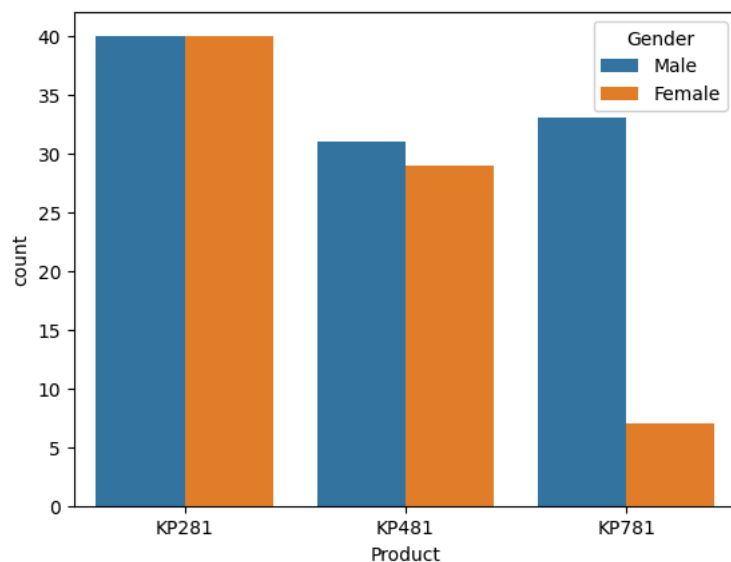
```
Mean Income 53719.57777777778
Median Income 50596.5
```

```
from itertools import product
# Bivariate Analysis
# relationship of 2 variables
"""
- Does income affect the treadmill purchase?
- Does the age influences miles run per week?
- Do males and females buy different products?
```

```
Variable 1    Variable 2
Age           Miles
Gender        product
Income        product
"""
```

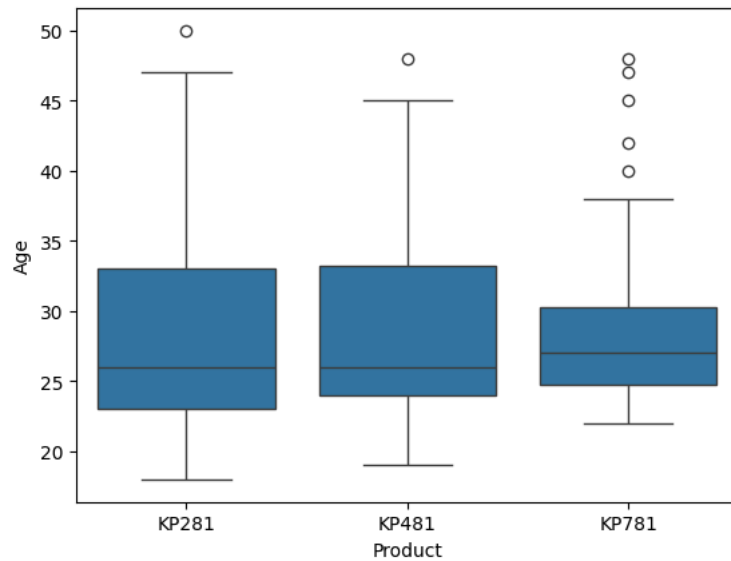
```
# product vs gender (categorical vs categorical)
```

```
sns.countplot(x="Product",hue="Gender",data=df)
plt.show()
```



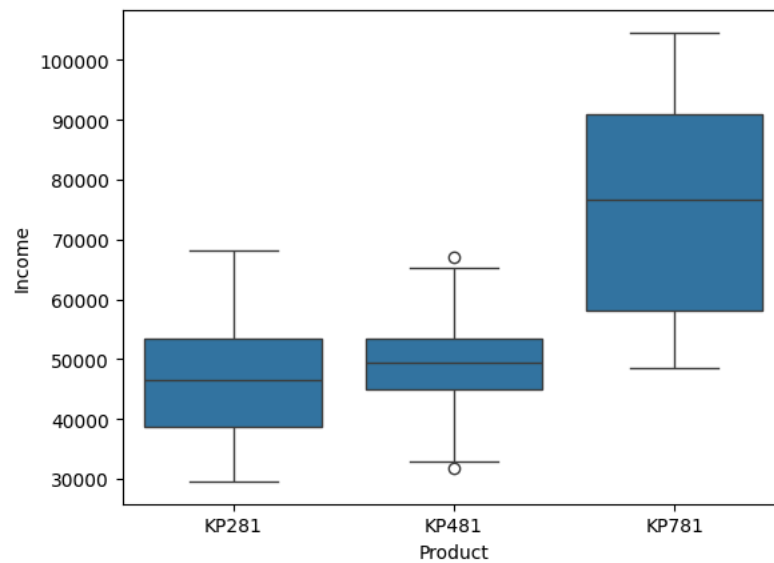
```
# Product vs Age ( Categorical vs Numerical)
```

```
sns.boxplot(x="Product",y="Age",data=df)  
plt.show()
```



```
# Product vs Income ( Categorical vs Numerical)
```

```
sns.boxplot(x="Product",y="Income",data=df)  
plt.show()
```

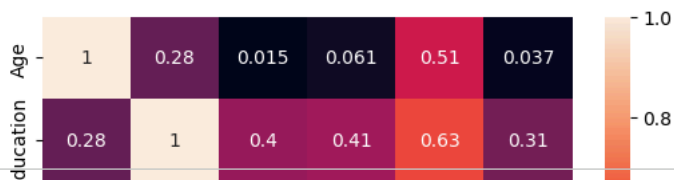


```
# Correlation - strength of relation between to variables
```

```
- Strength -1 to +1
```

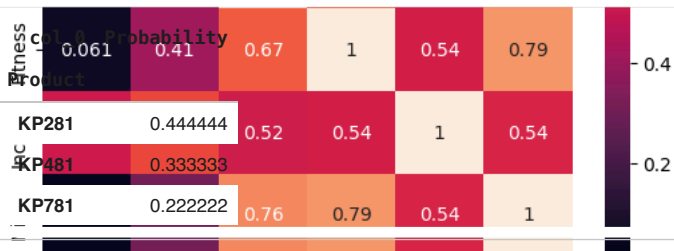
```
- correlation matrix - correlation coefficients between multiple values
```

```
corr=df.corr(numeric_only=True)  
sns.heatmap(corr, annot=True)  
plt.show()
```



Marginal Probability - single event

```
pd.crosstab(index=df["Product"], columns="Probability", normalize=True)
```



```
pd.crosstab(df["Gender"], df["Product"], normalize="index")
```

Product	KP281	KP481	KP781
Gender			
Female	0.526316	0.381579	0.092105
Male	0.384615	0.298077	0.317308

Customer profiling - understanding typical characteristics of a customer group

- What is the income level?
- How fit they are?
- How frequently they use the treadmill?

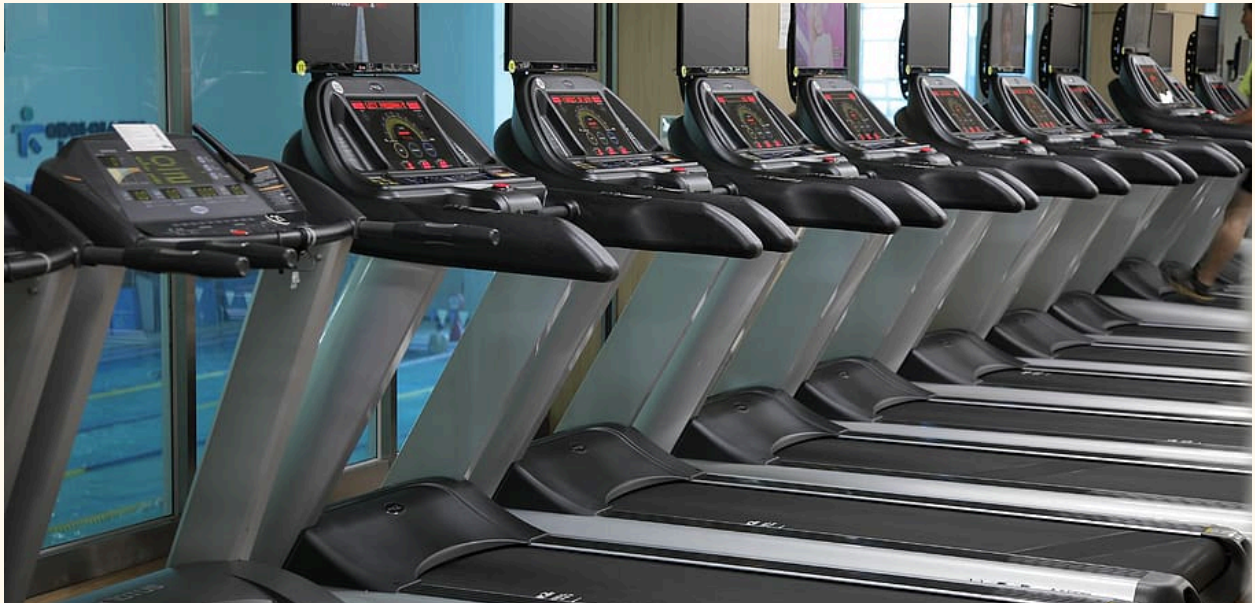
```
df.groupby("Product")[["Age", "Income", "Fitness", "Miles", "Usage"]].mean()
```

	Age	Income	Fitness	Miles	Usage
Product					
KP281	28.55	46418.025	2.9625	82.787500	3.087500
KP481	28.90	48973.650	2.9000	87.933333	3.066667
KP781	29.10	75441.575	4.6250	166.900000	4.775000

AEROFIT

a business case study

By Sourashis Sarkar



INTRODUCTION

Aerofit is a leading brand in the field of fitness equipment. The market research team at AeroFit wants to identify the characteristics of the target audience for each type of treadmill offered by the company, to provide a better recommendation of the treadmills to the new customers.

This is the analysis document of what I've researched. It has all the supporting information and graphs for a better understanding.

All the codes can be found in the Google Colaboratory file [here](#).

BASIC INFORMATION ABOUT DATASET

Preliminary info about the features of the Dataset:

Product Purchased: KP281, KP481, or KP781

Age: In years

Gender: Male/Female

Education: In years

MaritalStatus: Single or partnered

Usage: The average number of times the customer plans to use the treadmill each week.

Income: Annual income (in \$)

Fitness: Self-rated fitness on a 1-to-5 scale, where 1 is the poor shape and 5 is the excellent

Miles: The average number of miles the customer expects to walk/run each week

Product Portfolio

The **KP281** is an **entry-level** treadmill that sells for **\$1,500**.

The **KP481** is for **mid-level** runners that sell for **\$1,750**.

The **KP781** treadmill has **advanced** features that sell for **\$2,500**.

NON-GRAPHICAL ANALYSIS

1. Basic Analysis of Dataset

- 180 datapoints exist, each consisting of 9 features.
- No null values are present in any of the columns.
- Changed the fitness column to Low, Medium, and High values.
- Except Product, Gender and MaritalStatus, all the data are of integer type.
- Product, Gender, MaritalStatus, and Fitness are all categorical values.
- Age, Education, Usage, Income and Miles are numeric values.

2. Analysis of Categorical Variables

- Product
 - It consists of three values, “KP281”, “KP481” and “KP781”.
 - The most frequent product in the dataset is “KP281”, followed by “KP481” and “KP781”.
- Gender
 - It consists of two values, “Male” and “Female”.
 - It can be concluded from the dataset that Males tend to buy the treadmill more than the Females.
- Marital Status
 - It consists of two values, “Single” and “Partnered”.
 - It can be concluded from the dataset that married people tend to buy the treadmill more than the unmarried ones.
- Fitness
 - It consists of three values, “Low”, “Medium” and “High”.
 - The most frequent fitness in the dataset is “Medium”, followed by “High” and “Low”.

3. Analysis of Continuous Variables

- Age (in years)
 - Values range from 18 to 50.
 - Median value is 24.
 - The mean age is 28.7.
- Education (in years)
 - Values range from 12 to 21.
 - Median value is 16.
 - The mean of years of education is 15.57.

- Usage
 - Values range from 1 to 7.
 - Median value is 3.
 - Mean value is 3.4.
- Income (in \$)
 - Values range from \$29,562 to \$104,581.
 - Median value is \$50596.5.
 - Mean Income is \$53719.57.
- Miles
 - Values range from 21 to 360.
 - Median value is 94.
 - Mean value is 103.19.

MARGINAL PROBABILITY

Product		Fitness	
KP281	0.45	High	0.54
KP481	0.33	Medium	0.30
KP781	0.22	Low	0.16
Gender		Marital Status	
Male	0.58	Single	0.59
Female	0.42	Partnered	0.41

CONDITIONAL PROBABILITY

Probability of buying a product when certain conditions are given:

Conditions	Gender		Fitness			Marital Status	
Product	Male	Female	High	Medium	Low	Single	Partnered
KP281	0.38	0.53	0.20	0.56	0.54	0.44	0.45
KP481	0.30	0.38	0.15	0.40	0.46	0.33	0.34
KP781	0.32	0.09	0.65	0.04	0.00	0.23	0.21

Probability of buying a product given age:

	Age	(17, 20]	(20, 25]	(25, 30]	(30, 35]	(35, 40]	(40, 45]	(45, 51]
Product								
KP281		0.6	0.405797	0.512195	0.34375	0.500	0.500000	0.500000
KP481		0.4	0.347826	0.170732	0.53125	0.375	0.166667	0.166667
KP781		0.0	0.246377	0.317073	0.12500	0.125	0.333333	0.333333

As it is tiring to manually code every possibility, I've made a code cell which can be used to directly calculate probabilities based on the conditions. Anyone can directly access the cell from [here](#).

(Note: Run the whole colab notebook once and then rerun this code cell for this to work properly)

PROBABILITY OF CUSTOMERS BASED ON CONDITIONS

☐ Product

KP281

☐ Age

26 – 42

☐ Gender

Male

☐ Education

14 – 18

☐ Marital Status

Partnered

☐ Usage

3 – 5

☒ Fitness

Medium

☐ Income

48316 – 858

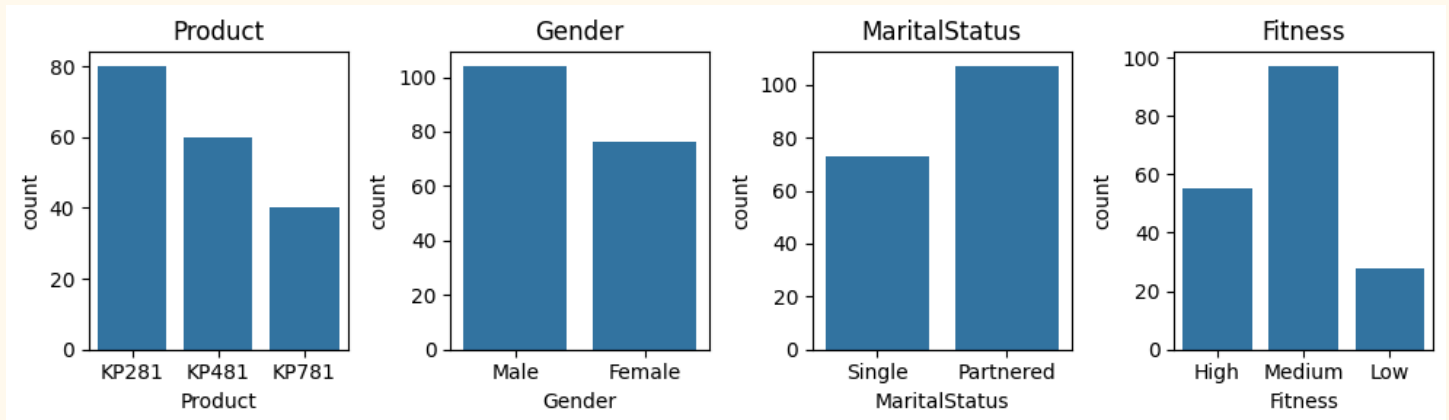
☐ Miles

105 – 275

Probability: 0.54

UNIVARIATE GRAPHICAL ANALYSIS

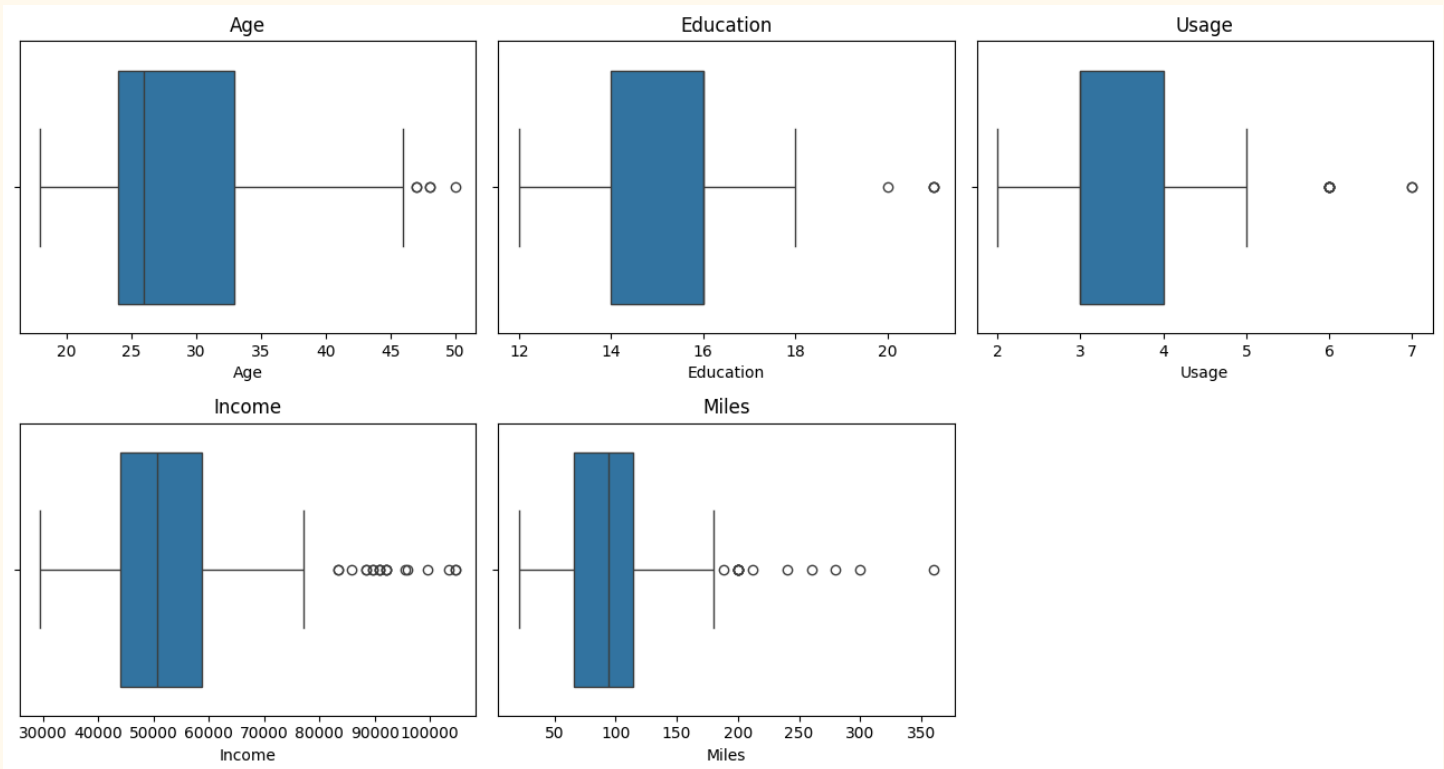
1. Countplot of all the categorical variables



Observations

- "KP281" is the most sold product, followed by "KP481" and "KP781".
- A male customer has more chances of buying the treadmill than females.
- A customer who is married has more chances of buying the treadmill than the unmarried ones.
- Customers with "Medium" fitness are more common, followed by "High" and "Low".

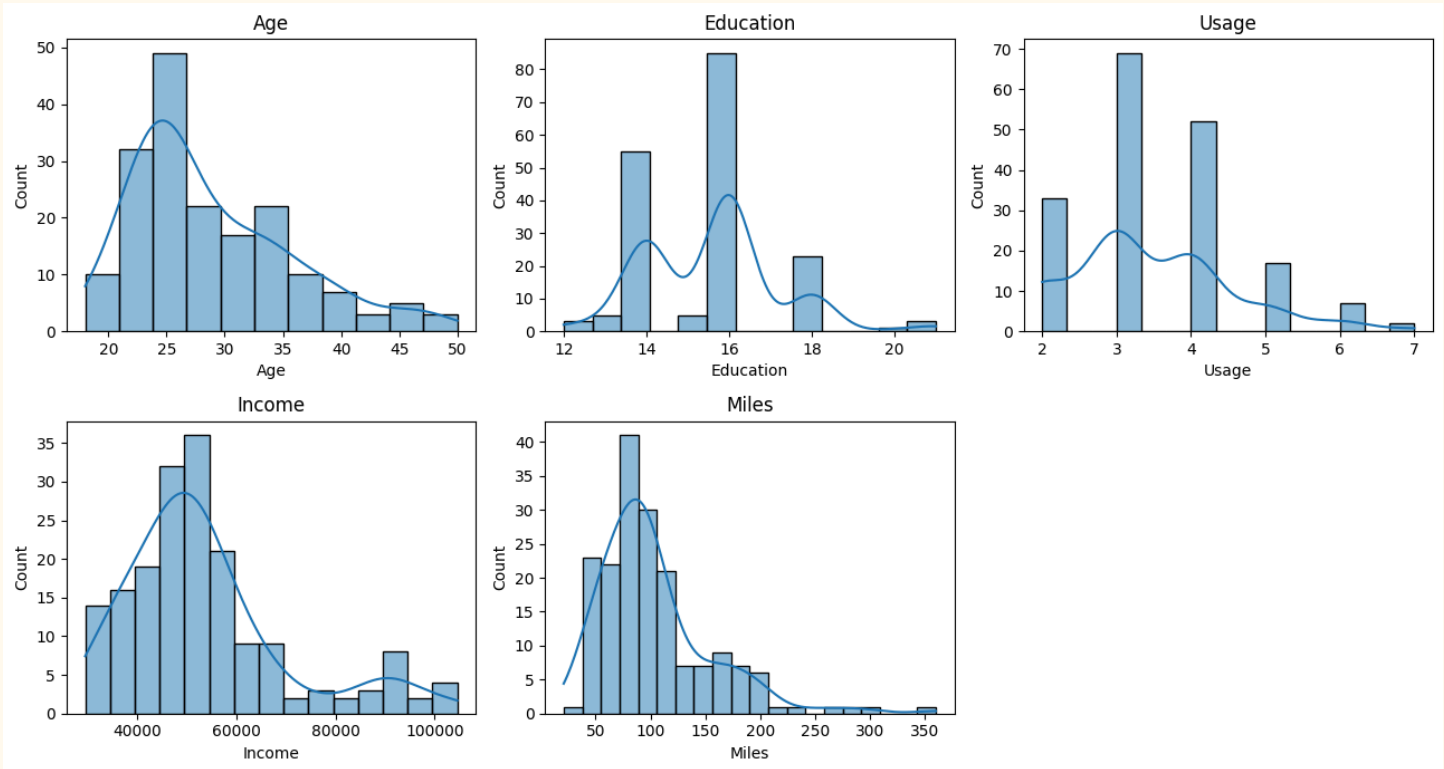
2. Boxplot of all the continuous variables



Observation

- Age is skewed to the left.
- Age has 5 outliers.
- Education has 4 outliers.
- Outliers of Usage: 9
- Outliers of Income: 19
- Outliers of Miles: 13

3. Histogram of all the continuous variables



Observations

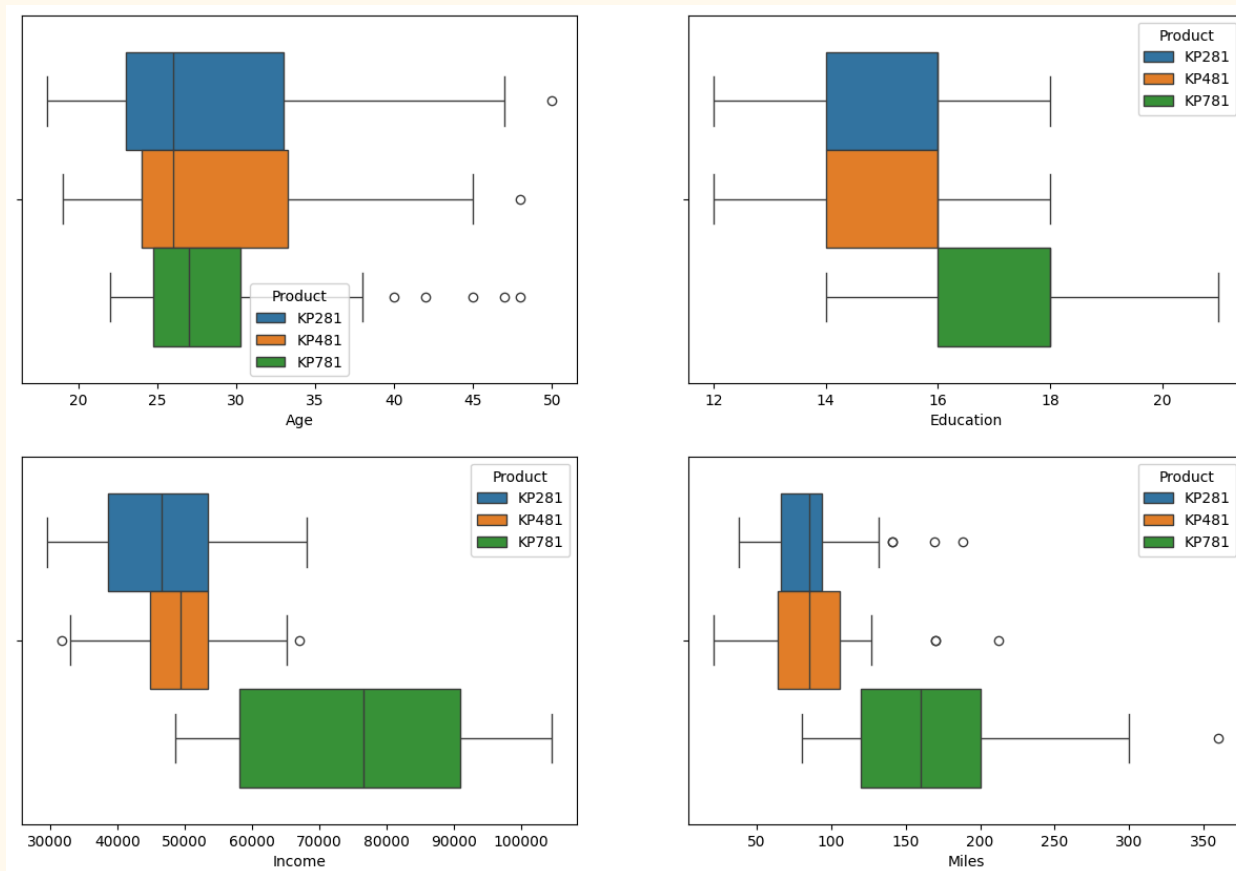
Maximum customers who buy the treadmill

- are in the ages of 20 to 35.
- have 14 to 18 years of education.
- plans to use the treadmill at least 3 times a week.
- considers themselves at least 3 on the fitness scale.
- earns in the range of \$20k to \$75k.
- expects to run 50-200 miles in a week.

Also, the Age and Miles do not follow exactly normal distribution, but it can be a log normal distribution with its tail stretched towards the right, whereas the others do not due to multiple peaks in the graph.

BIVARIATE GRAPHICAL ANALYSIS

1. Boxplots of Age, Education, Income and Miles based on different Products



Observations

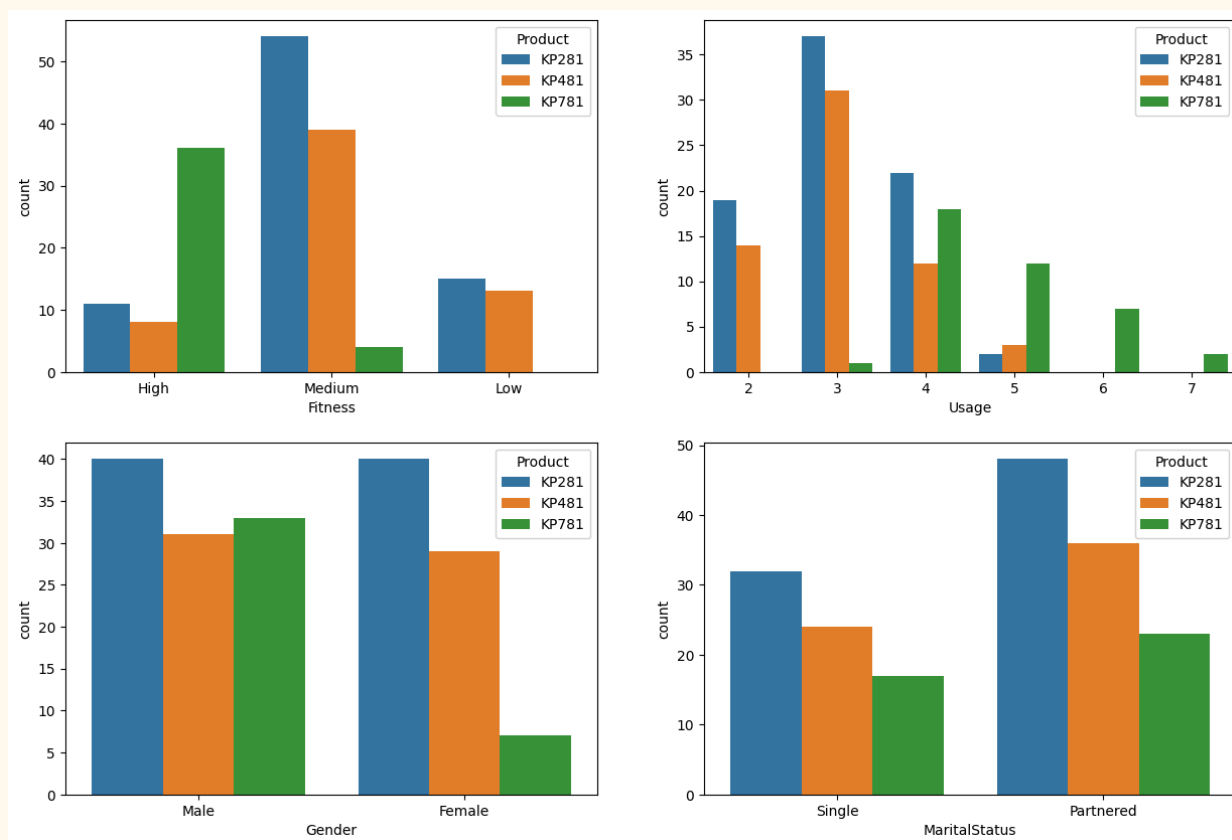
- Age
 - The boxplots for KP281 and KP481 show that the data is spread out over a larger age span. But the boxplot for KP781 is more clustered in nature.
 - The medians for all the boxplot is skewed towards left, which indicates that more datapoints are concentrated towards higher age.
 - Both the KP281 and KP481 have 1 outlier, and for KP781 we have 5 outliers.

- Education
 - Both the KP281 and KP481 have their whiskers on 12 and 18, but the whiskers of KP781 lie on 14 and 21. Hence, it can be inferred that a person with more education has an increased chance of buying KP781.
 - No outliers present.

- Income
 - The KP781 has a much larger spread than KP281 and KP481.
 - It can be inferred from the boxplot that people with higher income will have an increased chance to buy the KP781.
 - The KP481 plot shows 2 outliers, and the rest has no outliers present.

- Miles
 - The datapoints of KP281 are very clustered, followed by KP481. KP781 has much more spread than the other two.
 - It can be inferred that the people who expect to cover more miles per week have a greater chance of buying the KP781.
 - KP281 has 3 outliers, KP481 has 2 outliers, and KP781 has only 1 outlier.

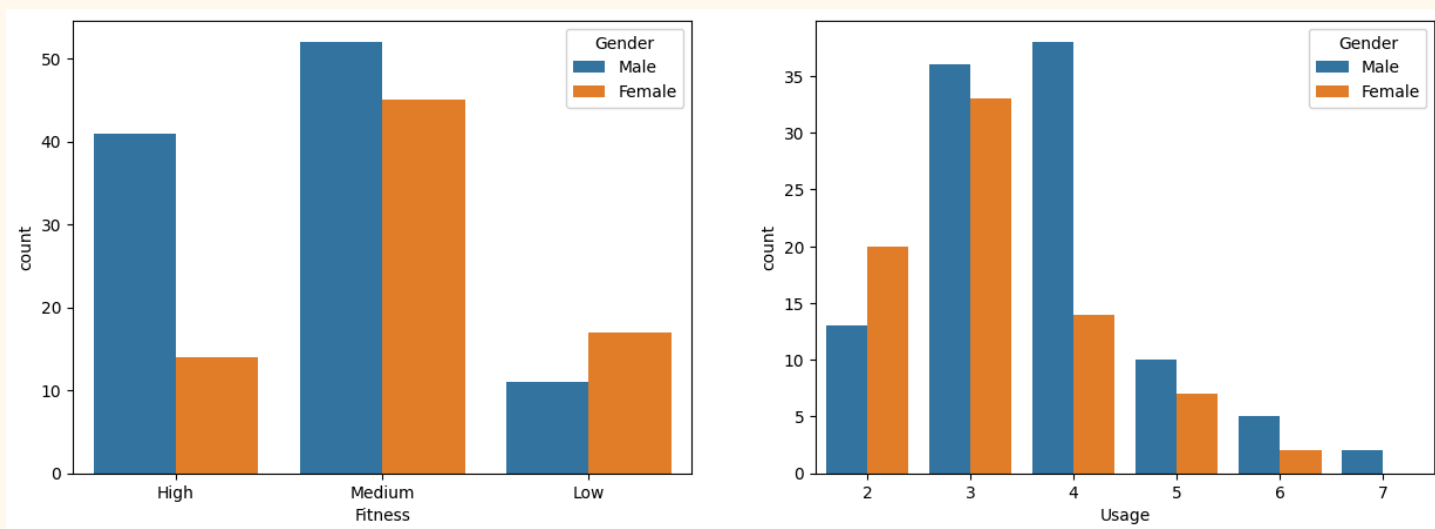
2. Barplots of Fitness, Usage, Gender, and Marital Status based on different Products



Observations

- Fitness
 - It can be inferred from the graph that a person who has medium or high fitness has a higher chance of buying a treadmill.
 - People who have a fitness high tend to buy the KP781 more.
- Usage
 - It can be inferred that generally the customers who buy the high end product (KP781) plans on using the treadmill at least 4 times a week.
 - Customers who plan on using the treadmill at most 4 times a week have the tendency to buy the low end or the mid range treadmill.
- Gender
 - In males, the high end treadmill is a little bit more favored than the mid range.
 - Women have a very low ratio of buying the high end treadmill.
- Marital Status
 - Overall, a married person has higher chances to buy a treadmill.

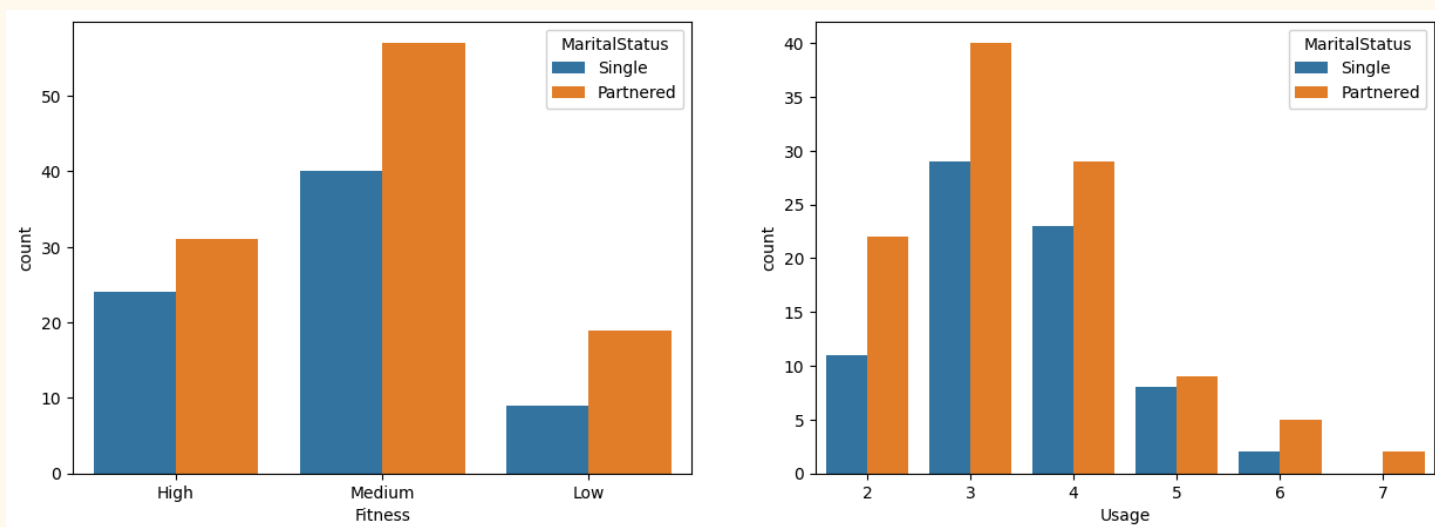
3. Barplots of Fitness and Usage based on Gender



Observations

- Overall, males show higher fitness and usage rate over females.

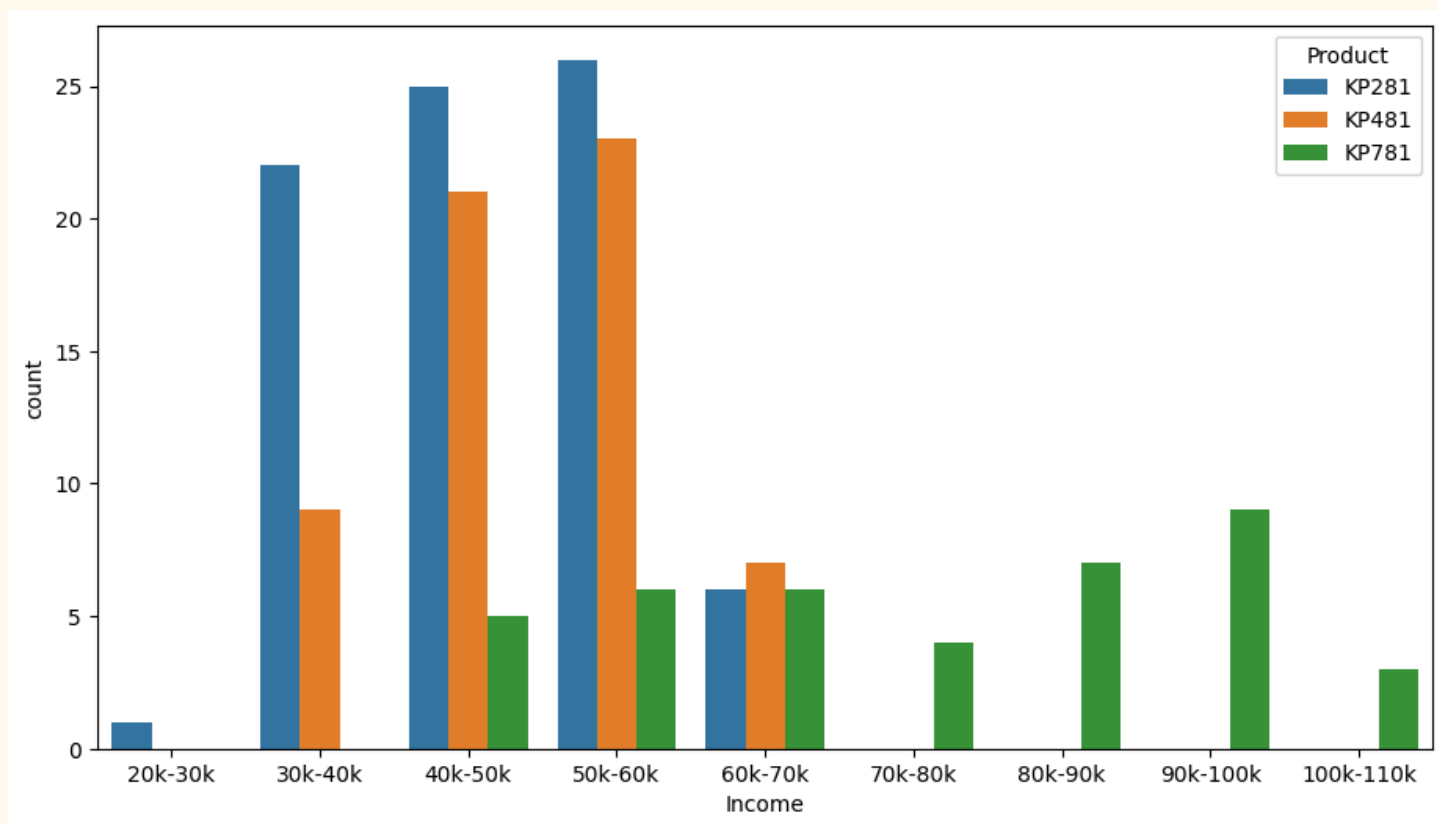
4. Barplots of Fitness and Usage based on Marital Status



Observations

- Overall, married people show higher fitness and usage rate over unmarried ones.

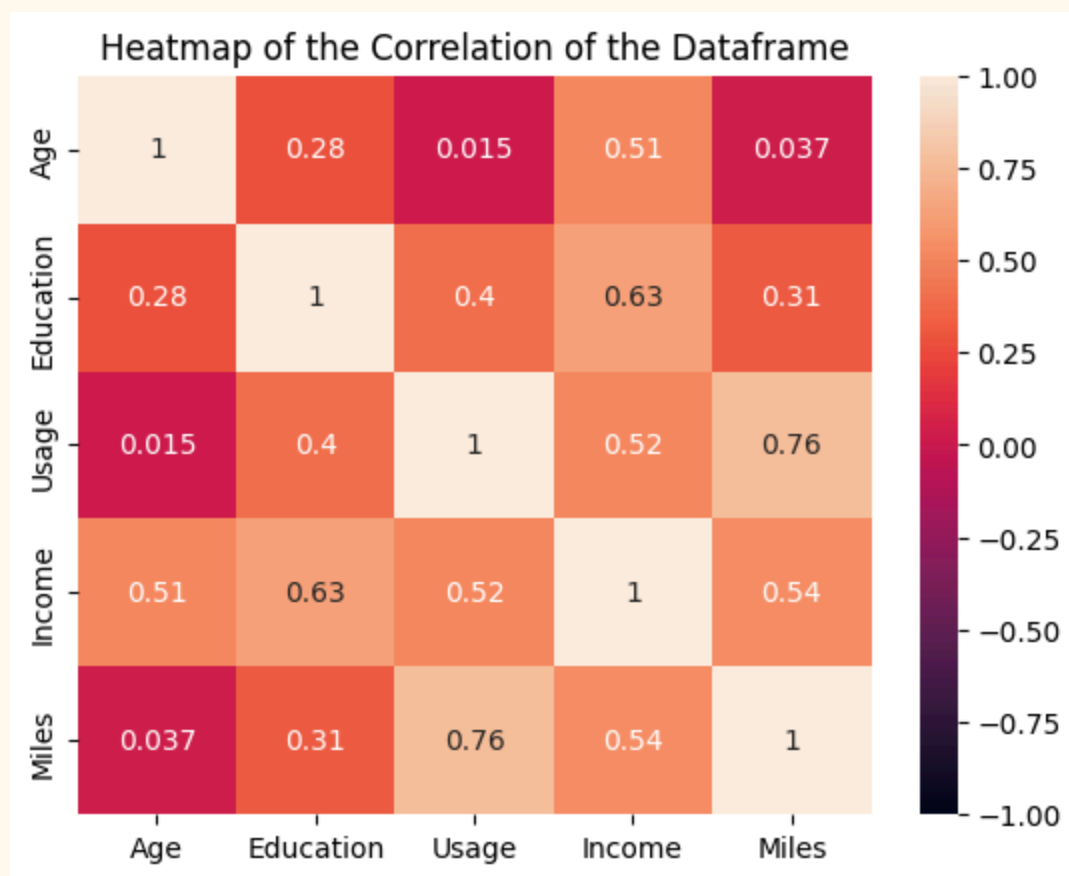
5. Countplot of Income Range based on Product



Observations

- Below \$60k, KP281 is the most popular.
- For \$60k-\$70k, KP481 is the most popular.
- Above \$70k, KP781 is the only sold item.
- High Income people prefer the high end model.
- Low Income people prefer the low end model.

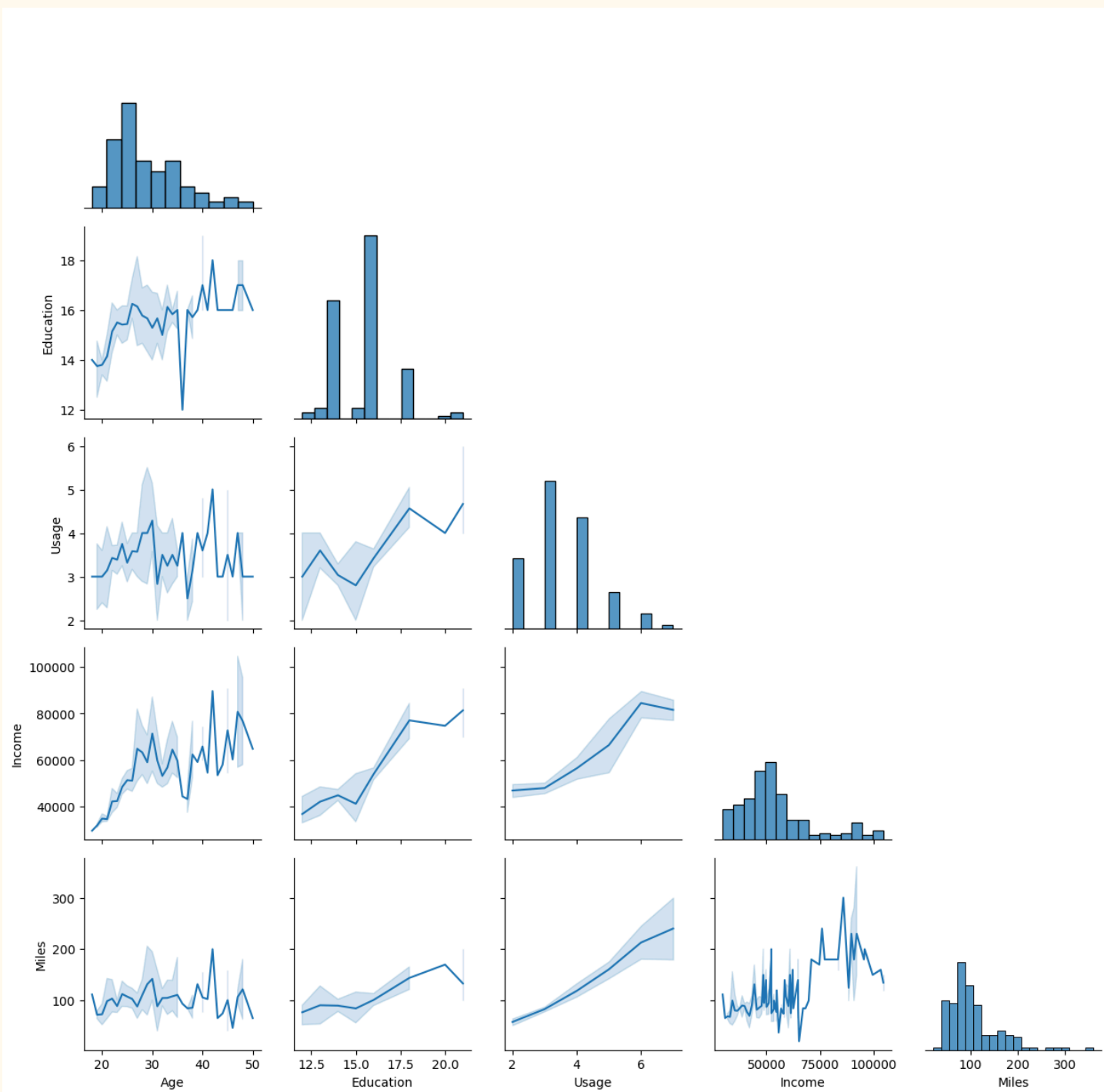
6. Finding correlation in the dataset from Heatmap and Pairplot



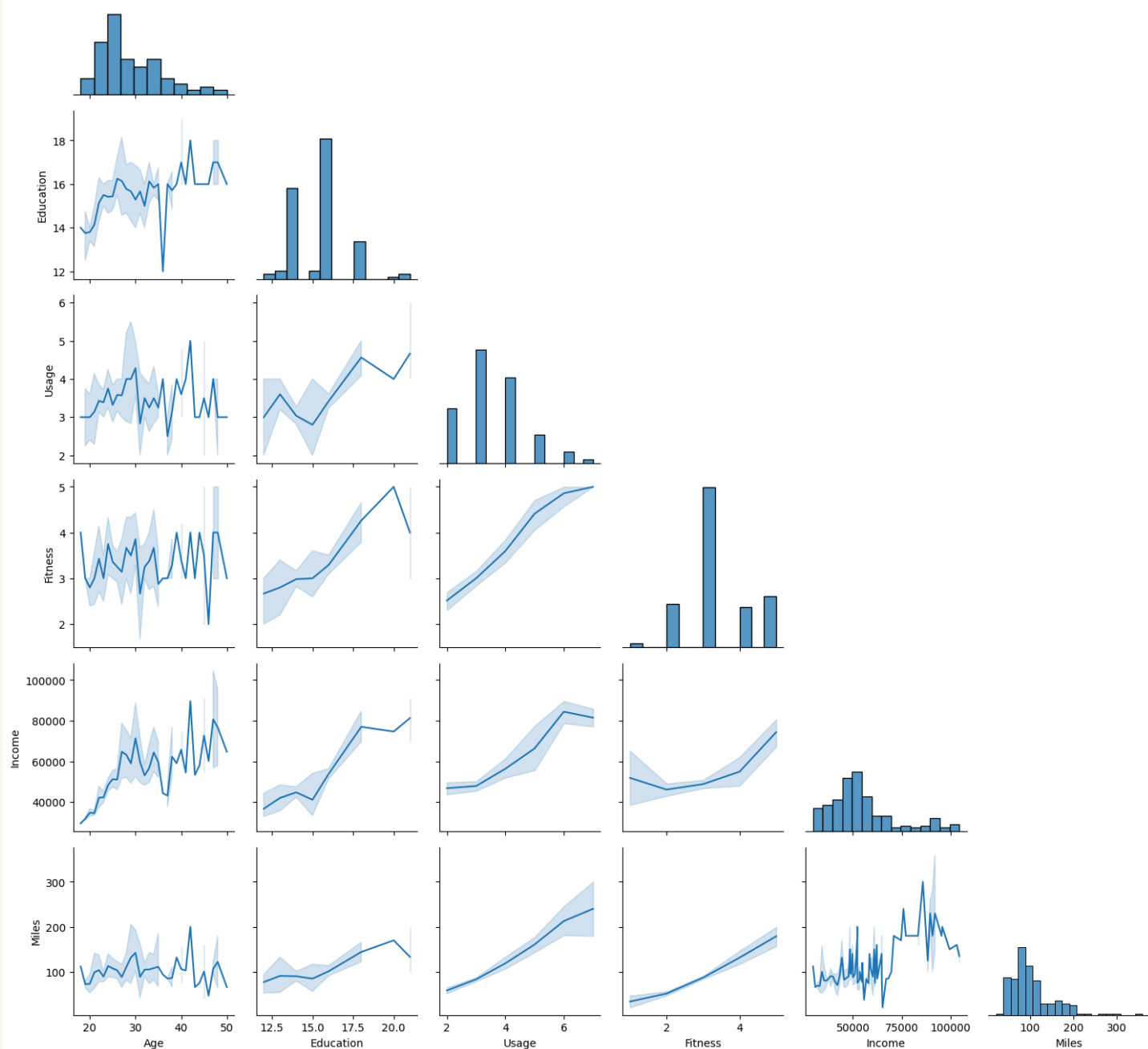
Observations

- Miles and Usage have the highest positive correlation of 0.76.
- Miles and Income have a positive correlation of 0.54.
- Usage and Income have a positive correlation of 0.52.
- There is no negative correlation in this data.

The correlation can be observed in the pairplots also.



Pairplot (with regplot)



Custom Pairplot (with lineplot)

CUSTOMER PROFILING

(based on median values)

1. Customer Profile for KP281

Product Purchased: *KP281*

Age: *26 years*

Gender: *Male*

Education: *16 years*

Fitness: *Medium*

Usage: *3 times per week*

Income: *\$46,617 annually*

Miles: *85 miles per week*

2. Customer Profile for KP481

Product Purchased: *KP481*

Age: *26 years*

Gender: *Male*

Education: *16 years*

Fitness: *Medium*

Usage: *3 times per week*

Income: *\$49,459 annually*

Miles: *85 miles per week*

3. Customer Profile for KP781

Product Purchased: *KP281*

Age: *27 years*

Gender: *Male*

Education: *18 years*

Fitness: *High*

Usage: *5 times per week*

Income: *\$76,568 annually*

Miles: *160 miles per week*

RECOMMENDATIONS

- It is recommended to have a good stock of KP281 as it is more in demand than the other two.
- To boost the sale of midrange and high level treadmills, there can be some discounts planned for special occasions or for a particular range of time.
- As female to male customer ratio is very less, especially in midrange and high level treadmills, we can separately focus on creating advertisements targeted to women and can even have discounts for them.
- Also, the single to married customer ratio is slightly lower, and to fix that, we can tailor advertisements around them.
- We can convince people to increase their usage as it can lead to them buying a midrange or high level treadmill.
- We can spread awareness about how important our physical health is to attract customers.
- We can attract customers with high income and can give special benefits if they purchase the high level model.
- We can promote the high end treadmill to high income and high educated people.

Feature

Meaning

Product

Treadmill

Age

Customer's age

Gender

Customer's gender

Education

Years of education

Marital Status

Single / Partnered

Usage

times per week

Income

Yearly income

Fitness

Self rating (1-5)

Miles

Miles per week

	V1	V2	V3	V4	V5
V1	1		0.2		
V2		1			
V3			1		
V4				1	
V5					1

+1

+ve relation perfect

0.7 to 0.9

Strong +ve

0.4 to 0.6

Moderate

0.1 to 0.3

Weak

0
Negative

No relation
Opposite relation

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
url="https://d2beiqkhq929f0.cloudfront.net/public_assets/assets/000/001/125/original/aerofit_treadmill.csv?163999"
df=pd.read_csv(url)
```

```
# df.head() -> shows n no of rows of the table
df.head() # 5 by default
```

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
0	KP281	18	Male	14	Single	3	4	29562	112
1	KP281	19	Male	15	Single	2	3	31836	75
2	KP281	19	Female	14	Partnered	4	3	30699	66
3	KP281	19	Male	12	Single	3	3	32973	85
4	KP281	20	Male	13	Partnered	4	2	35247	47

```
# df.tail() # last 5 default rows
df.tail()
```

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
175	KP781	40	Male	21	Single	6	5	83416	200
176	KP781	42	Male	18	Single	5	4	89641	200
177	KP781	45	Male	16	Single	5	5	90886	160
178	KP781	47	Male	18	Partnered	4	5	104581	120
179	KP781	48	Male	18	Partnered	4	5	95508	180

```
# check dataset structure
df.shape # 180 rows, 9 cols
```

```
(180, 9)
```

```
df.info() # datatypes
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 180 entries, 0 to 179
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Product                180 non-null   object
1   Age                   180 non-null   int64
2   Gender                180 non-null   object
3   Education             180 non-null   int64
4   MaritalStatus         180 non-null   object
5   Usage                 180 non-null   int64
6   Fitness               180 non-null   int64
7   Income                180 non-null   int64
8   Miles                 180 non-null   int64
dtypes: int64(6), object(3)
memory usage: 12.8+ KB
```

```
df.describe() # stat summary
```

	Age	Education	Usage	Fitness	Income	Miles
count	180.000000	180.000000	180.000000	180.000000	180.000000	180.000000
mean	28.788889	15.572222	3.455556	3.311111	53719.577778	103.194444
std	6.943498	1.617055	1.084797	0.958869	16506.684226	51.863605
min	18.000000	12.000000	2.000000	1.000000	29562.000000	21.000000
25%	24.000000	14.000000	3.000000	3.000000	44058.750000	66.000000
50%	26.000000	16.000000	3.000000	3.000000	50596.500000	94.000000
75%	33.000000	16.000000	4.000000	4.000000	58668.000000	114.750000
max	50.000000	21.000000	7.000000	5.000000	104581.000000	360.000000

```
# missing values
df.isnull().sum() # True False table
```

	0
Product	0
Age	0
Gender	0
Education	0
MaritalStatus	0
Usage	0
Fitness	0
Income	0
Miles	0

dtype: int64

```
# which treadmill was bought the most? / which is the most popular product
df["Product"].value_counts()
```

```
# KP281
```

	count
Product	
KP281	80
KP481	60
KP781	40

dtype: int64

```
df["Gender"].value_counts()
```

```
# Who buys more treadmills? Males buy more treadmills
```

	count
Gender	
Male	104
Female	76

dtype: int64

```
df["MaritalStatus"].value_counts()
```

	count
MaritalStatus	
Partnered	107
Single	73

dtype: int64

```
# Univariate Analysis
```

```
# Average age of customers?
# What is the min and max age or cutoemrs?
# Which age group buys the most treadmills?
```

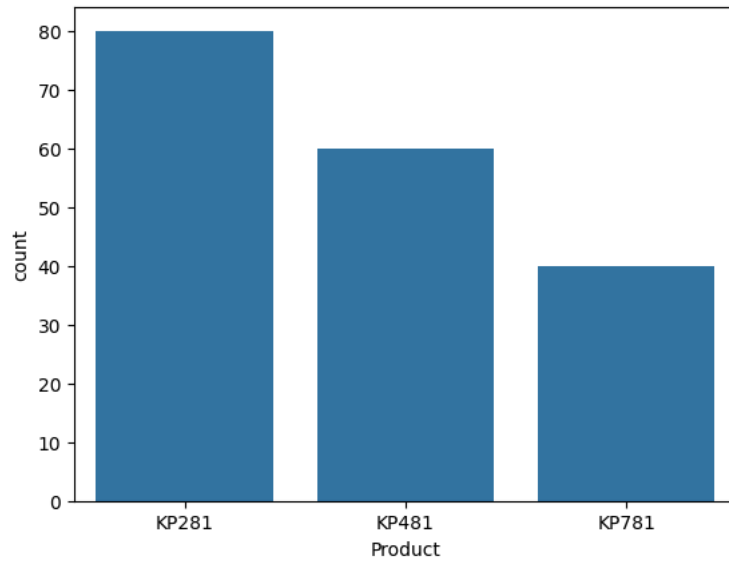
```
# Product distribution
```

```
# sns.countplot(x="column", data=df)
```

```
sns.countplot(x="Product",data=df)
```

```
# KP281 - most popular -> entry level product
```

<Axes: xlabel='Product', ylabel='count'>

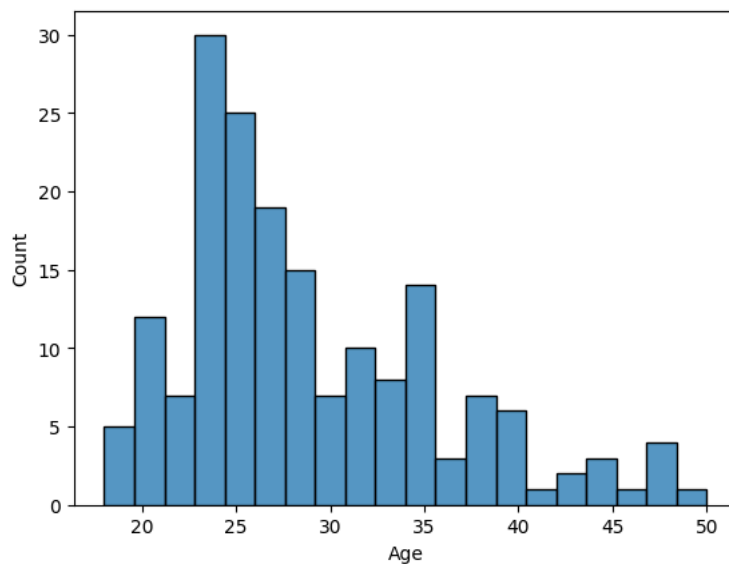


age distribution

```
sns.histplot(df["Age"],bins=20)
```

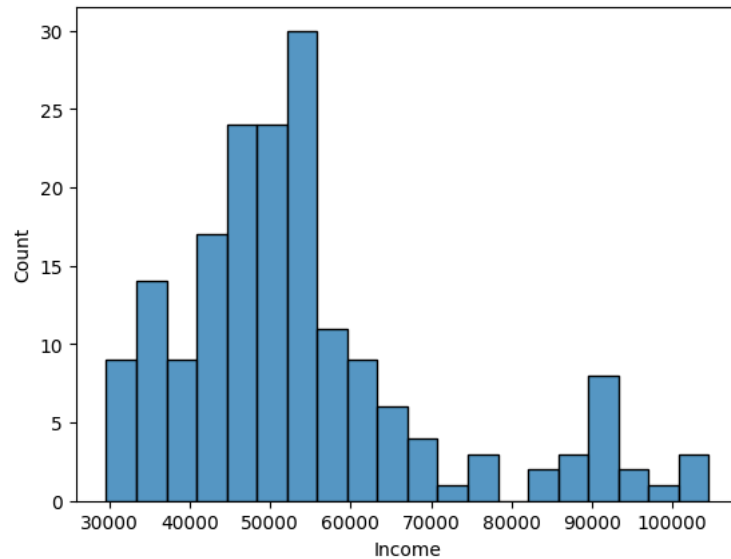
most customers who buy our treadmills are of the age group 25 – 25

<Axes: xlabel='Age', ylabel='Count'>



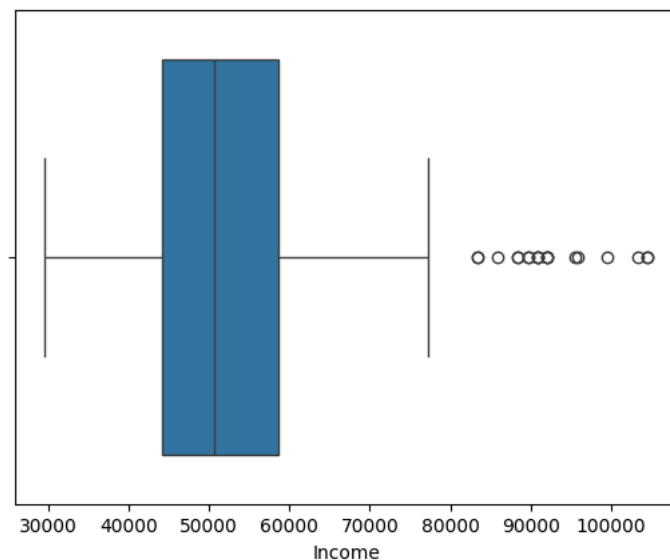
```
sns.histplot(df["Income"],bins=20)
```

<Axes: xlabel='Income', ylabel='Count'>



```
# outlier detection
```

```
sns.boxplot(x=df["Income"])
plt.show()
```



```
print("Mean Income", df["Income"].mean())
print("Median Income", df["Income"].median())
```

```
Mean Income 53719.57777777778
Median Income 50596.5
```

```
# Bivariate Analysis
```

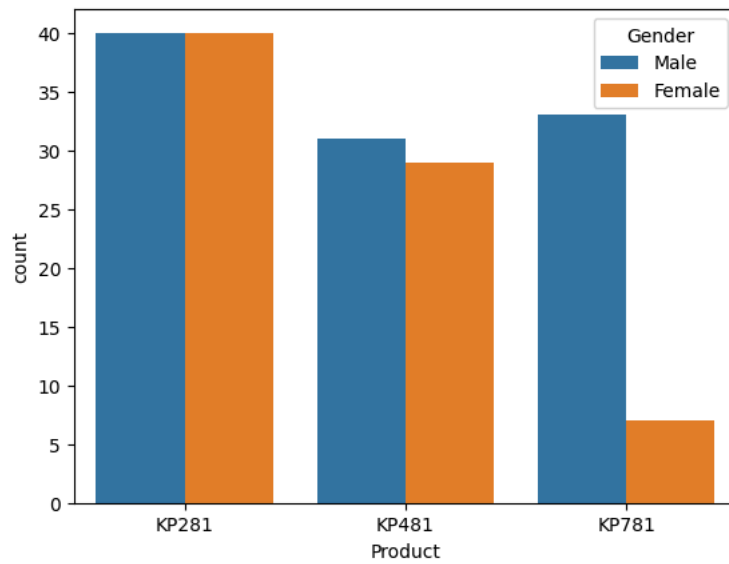
```
# Does income affect the product will we purchase? - var 1 Income, var 2 Product
```

```
# Does age influence miles run per week? - var 1 - Age, var 2 - Miles
```

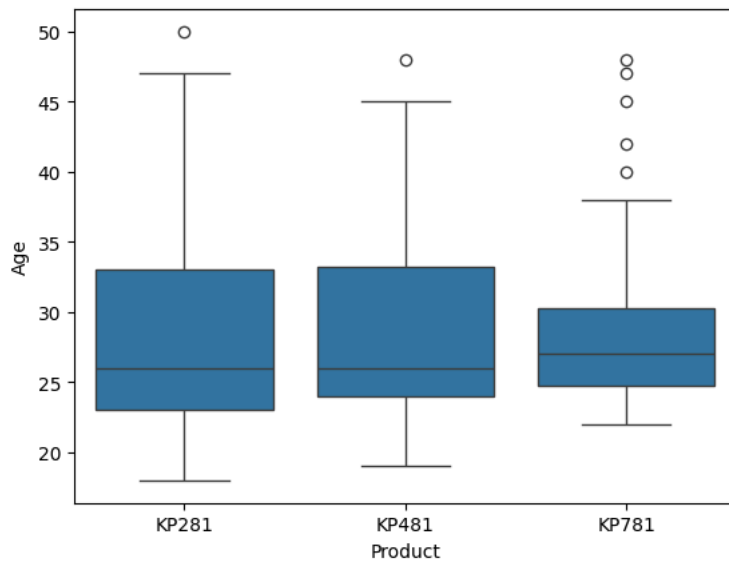
```
# Do males and females buy different kind of products? - Gender - product
```

```
# Product VS Gender (Categorical vs Categorical)
```

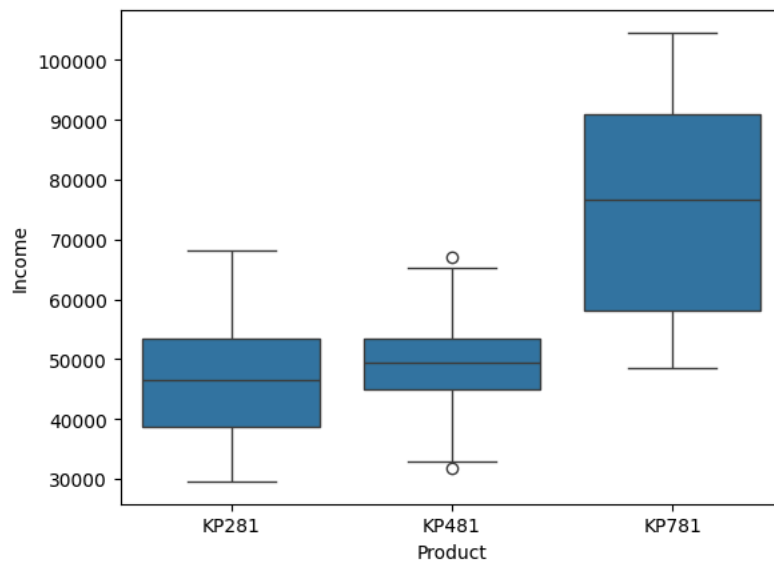
```
sns.countplot(x="Product",hue="Gender",data=df)
plt.show()
```

```
# Product vs Age (Product vs age)
sns.boxplot(x="Product",y="Age",data=df)
plt.show()
```



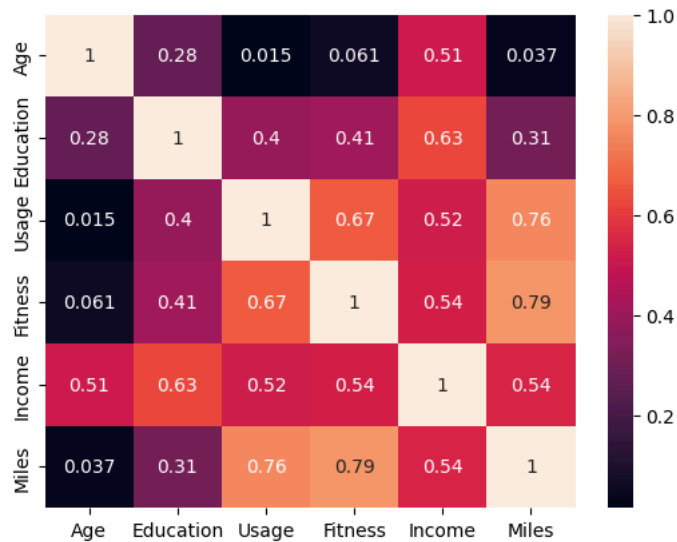
```
# Product vs Income
sns.boxplot(x="Product",y="Income",data=df)
plt.show()
```



```
# COrrealtion Analysis
"""
Measures the strength and direction of relationship between 2 variables
Numerical values

Correlation matirx - table showed correlation coefficients
"""
```

```
corr=df.corr(numeric_only=True)
sns.heatmap(corr,annot=True)
plt.show()
```



```
# Probability of single event - marginal prob
pd.crosstab(index=df["Product"],columns="Probability", normalize=True)
```

```
col_0 Probability
Product
KP281      0.444444
KP481      0.333333
KP781      0.222222
```

```
# Customer Profiling
- What is the average agr group?
- What is the income level?
- How fit are they?
- How many miles can they run?
- How frequently they use the treadmill?
```